

A Project Report

On

**“A Detailed Analysis of Different Types of Federated Learning
Models for Crop Disease Detection”**

Submitted to

Maulana Azad National Institute of Technology, Bhopal



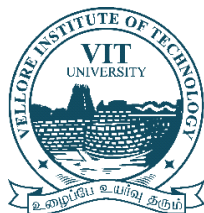
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CERTIFICATE

This is certified that the project entitled “A Detailed Analysis of Different Types of Federated Learning Models for Crop Disease Detection” was carried out under my supervision.

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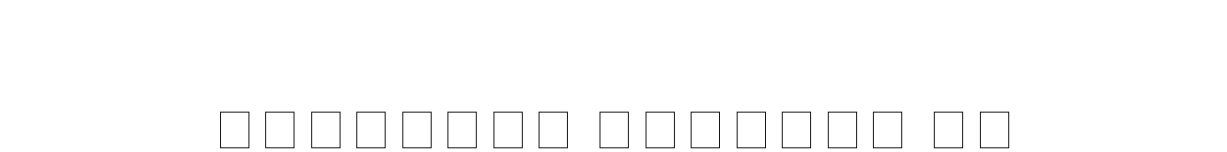


Abstract

Machine learning (ML) techniques are used to predict crop yield, early detection of crop diseases, estimation of crop damage by weather parameters such as atmospheric temperature and humidity, assessment of disease severity, and detection of soil nutrition, water quality management, crop monitoring, livestock monitoring. Researchers have developed various solutions based on artificial neural networks (ANN) to simplify agricultural tasks. In 2016, Google introduced FL, an emerging approach that addresses this limitation by allowing knowledge sharing while maintaining privacy and reducing costs. FL allows devices to collectively learn a shared model while keeping learning data stored on devices, thus separating machine learning functionality from the need for centralized data storage. FL is also beneficial in resource-constrained situations, such as mobile devices, IoT devices, or networks with restricted bandwidth, because it tries to be both effective and scalable. Federated Learning (FL) is an emerging technique that offers significant potential to enhance smart agriculture by enabling collaborative model training across distributed data sources while preserving data privacy. This research provides a comprehensive overview of the integration of FL within smart agriculture, emphasizing its role in addressing key challenges, such as data privacy, security, scalability, and data heterogeneity. The research project creates a decentralized deep learning framework which maintains privacy during plant disease detection through federated learning. The study employs a massive dataset from Plant Village containing approximately 70,000 labelled images which show healthy and diseased leaves across 9 plant species including tomato, apple, bell pepper, peach and strawberry. A ResNet-50 convolutional network is used to adapt to this dataset. The study evaluates three federated optimization schemes FedAvg, FedAdam, and FedProx during multiple-client testing with each client specializing in one specific crop species. The global model receives training through repeated communication rounds that involve synchronized updates on each client. The research provides detection performance results together with precision and recall metrics as well as F1 scores for test data. The experiments conducted with 4 to 9 clients and 32 batch sizes and learning rate $1e-4$ and local epochs between 5 and 10 per round demonstrate that the federated ResNet-50 delivers excellent disease detection accuracy.

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other institute. I hereby declare that the facts mentioned above are true to the best of our knowledge. In the event of any unlikely discrepancy that may possibly occur, I will be the one to take responsibility.

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A Detailed Analysis of Different Types of Federated Learning Models for Crop Disease Detection

6. Introduction

6.1 Background

Agricultural practices have been revolutionized as a result of the convergence of technology and agriculture in recent years [1]. Agricultural productivity can be enhanced, resource usage optimized, and environmental impact mitigated through the use of cutting-edge technologies like Internet of Things (IoT) [2], artificial intelligence (AI) [3], and big data analytics. A promising paradigm among these technologies is Federated Learning (FL)[4], which has the potential to transform agriculture.

Federated Learning takes a decentralized approach [5], where models are trained locally on distributed datasets from multiple devices or locations, without centralized data aggregation. A fundamental challenge faced by the agriculture sector is the dilemma between data sharing and privacy. A vast amount of information is available to farmers and agricultural stakeholders, including crop yields, soil conditions, weather patterns, and pest incidences. Despite this, concerns about data privacy and ownership have prevented the sharing of this data for collaborative analysis. As an application of Federated Learning in agriculture, Federated Learning offers a solution that reconciles data privacy with collaborative model training. FL preserves the privacy of sensitive agricultural data by allowing machine learning models to be trained directly on local data on farmers' devices or servers, while still leveraging the collective insights derived from distributed datasets.

This decentralized approach not only protects unique information from farmers, but also promotes cooperation and innovation across agroecosystems. Along with the FL, farmers can benefit from the collective intelligence of device networks. This leads to more accurate forecasts, improved crop yields and sustainable agricultural practices without affecting data protection [7]. This innovative technology could potentially change the agriculture industry by enabling efficient resource allocation, reducing environmental impacts, and ultimately ensuring nutritional security for the world's population.

6.1.1 Smart agriculture using federated learning

In an FL network, local nodes (end devices) send their updated weights to the server, which aggregates these weights and sends back the updated model weights for further training at the edge nodes. This process continues until convergence

is achieved. The architecture of FL in smart agriculture is depicted in Figure 2. The following steps outline the process.

- 1) Various sensors deployed in farms monitor and collect data related to agriculture such as soil conditions, weather, irrigation, light, temperature, positioning, and air quality
- 2) The collected data is then transmitted to the local database
- 3) The local sites train their respective model using the data in local database
- 4) Updates of locally trained models are uploaded to the central server
- 5) The collected updates from multiple clients aggregated using aggregation algorithms such as FedAvg, FedProx
- 6) The updated model weights are sent back to the clients

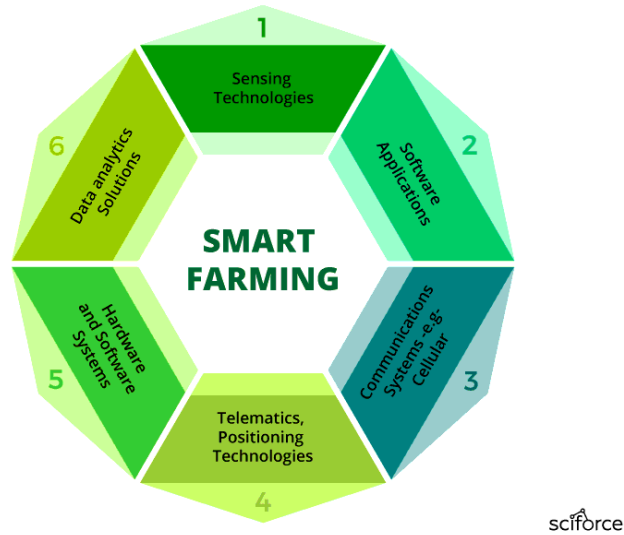


Fig 1. Smart Farmhouse

6.2 Purpose

Federated Learning helps collaborative studies and know how sharing inside the agricultural community. By allowing stakeholders to together educate device studying fashions on allotted datasets, FL promotes the trade of information and insights whilst respecting records privateness and possession rights. This collaborative technique fosters innovation and speeds up the adoption of records-pushed answers to cope with the complicated demanding situations dealing with present day agriculture. By allowing decentralized device studying on allotted datasets, FL empowers farmers and agricultural stakeholders to harness the collective intelligence embedded of their records whilst retaining records privateness and possession rights.

This approach safeguards sensitive information, such as farm locations or proprietary crop strains, while still benefiting from a large, combined dataset. Recent studies have started to explore federated CNNs for monitoring plant health.

Inspired by these achievements, we are exploring federated learning for single-crop disease detection using a much larger and more diverse dataset derived from PlantVillage. In our setup, each federated client holds images for a specific crop, such as

tomatoes or apples, and a global ResNet-50 model is utilized. As the agriculture region keeps to include virtual transformation, Federated Learning emerges as a key enabler of innovation, sustainability, and resilience in agriculture.

6.3 Research Motivation

The growing world population and the need for sustainable use of resources have made things harder for the agriculture industry. Although traditional farming methods have been important, they are not enough anymore to deal with problems like less farmland, unpredictable weather patterns, and more frequent crop diseases. Machine Learning (ML) has become a key solution in modern farming, helping in areas such as predicting harvests, spotting diseases early, and managing resources more accurately. But these ML tools work well only if they have access to large, varied, and accurate data sets. This often means collecting data from many places, which can create challenges with keeping data private and secure, as well as dealing with the difficulties of combining different types of data from various locations.

Farmers and agricultural stakeholders often hesitate to share their valuable data with a central system because of privacy worries. This data includes secret information about crops, exact farm locations, and specific farming methods. Their reluctance is due to real concerns about who owns the data, how it might be used, and the risk of losing a competitive edge. As a result, the full potential of AI working together in agriculture is not being reached. Most current AI systems require sending raw data to external servers, which creates security risks and makes the farming community less trusting. This setup also has trouble growing and working well in places with limited resources, like remote farms with poor internet or large networks of smart devices used in farming.

This research is fundamentally motivated by the critical need to reconcile the immense benefits of data-driven agricultural AI with the paramount importance of data privacy and security. The advent of Federated Learning (FL) presents a groundbreaking solution to this dilemma. FL allows for the collaborative training of robust machine learning models across decentralized datasets, where individual data remains securely stored on local devices. This paradigm shift not only safeguards sensitive information but also significantly reduces the computational and bandwidth overhead associated with transmitting large volumes of raw data to a central server. By fostering a decentralized learning environment, FL empowers

farmers to contribute to the collective intelligence of an AI system without compromising the confidentiality of their operations.

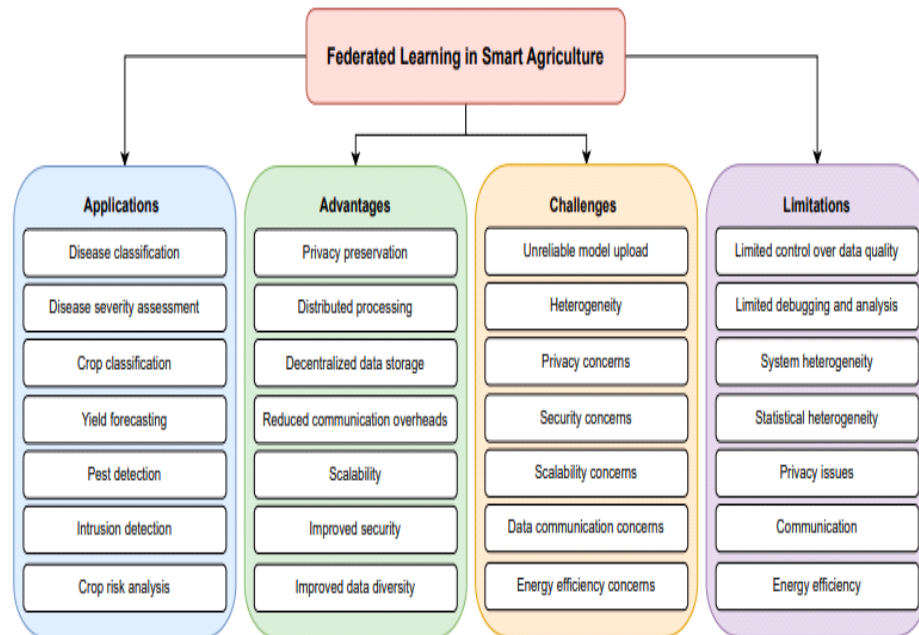


Fig 2. Federated Learning in Smart Agriculture

6.4 Significance of Research

The significance of this research is multi-faceted and has profound implications for the future of smart agriculture:

Enabling privacy-friendly teamwork: Federated Learning (FL) lets different parties in the agriculture industry work together to make machine learning models better, without ever sharing or exposing their actual data. Important details like where a farm is located or specific crop types remain safe on the devices used by each party, helping to overcome major challenges in sharing information.

Improving model performance and adaptability: Farms vary a lot in terms of conditions and challenges. By combining insights from many different farms—each with its own crops, weather, or disease issues—FL models, especially when using advanced techniques like FedAvg, FedProx, or FedAdam, learn to recognize patterns that work across different situations. This makes the models more accurate and reliable when used in real farming situations. **Saving on resources and data transfer:** With FL, only the model's updated settings are sent over the network, not the full data. This greatly cuts down on how much data is sent

and the processing power needed, which is especially helpful for farms with slow internet or devices like IoT tools and mobile phones.

Pioneering Privacy-Preserving AI in Agriculture: This study directly addresses the long-standing tension between data utility and data privacy in agriculture. By implementing a decentralized deep learning framework for plant disease detection using FL, we demonstrate a viable pathway for harnessing the power of AI while ensuring farmers retain complete control and ownership over their sensitive on-device image data. This is a crucial step towards building trust and encouraging wider adoption of AI solutions in a privacy-conscious agricultural landscape.

Enhancing Scalability and Resource Efficiency: The proposed FL framework circumvents the need for massive centralized data storage and computationally intensive server-side processing of raw data. This decentralized approach is particularly beneficial for resource-constrained environments prevalent in many agricultural settings, such as mobile devices, IoT sensors, and areas with limited network bandwidth. By keeping data localized, the framework reduces communication costs and optimizes resource utilization, making advanced AI accessible to a broader range of agricultural stakeholders.

Advancing Collaborative Intelligence and Knowledge Sharing: FL fosters a collaborative environment where agricultural organizations from different locations can work together to train a common model without sharing their actual data. This allows the pooling of varied knowledge from different local data sets, resulting in stronger and more general disease detection models that can work well with a variety of plants and growing conditions. This form of shared intelligence is key to speeding up innovation and spreading effective farming practices throughout the agricultural sector.

Empirical Validation of Federated Optimization Schemes: The study carefully tests the performance of three major federated learning methods—FedAvg, FedAdam, and FedProx—in real-world plant disease detection scenarios.

Using a large and varied dataset from PlantVillage and simulating situations where multiple clients specialize in different crops, the research provides important information on how well these methods handle differences in data and achieve consistent results in agricultural federated learning systems. A detailed comparison of precision, recall, and F1 scores gives useful direction for future use of these methods.

7. Literature Survey

Recent studies have shown the practical potential of FL in agriculture, particularly in the detection of plant diseases. The author Mamba et al. (2023) [1] implemented the Feder-50 model for classification of spinach leaf disease, achieving an impressive accuracy of up to 97% in several customers. Her work showed that the power of the central system could be consistent with the federation model, even when data were distributed across different farms.

Similarly, Kabbalah et al. (2023) [2] further confirmed that a model of data protection bonds applied to the PlantVillage dataset and that can be created that is comparable to diseases trained with aggregated data. These successful structures expand on the current research FL experiment using larger and more diverse data records. By using a 50 remain folding network grown on Imagenet and adapting to data records of 70,000 marked images from nine harvest types, the researchers have shown that feather models can achieve villains above 90°C when detecting plant disease. This is achieved while ensuring that sensitive image data maintains data protection without leaving the farmer's device.

Mehta et al. [3] used a lightweight convolutional neural network model to identify and categorize mango leaf diseases. The effectiveness of this lightweight model was evaluated in four different clients and five disease categories, including healthy, anthracnose, powdery mildew, leaf spot, and leaf curl. The researchers achieved a detection accuracy of 98% for these particular diseases.

In a study by Aggarwal et al. [4] the EfficientNetB3 model was used to detect rice diseases. The data set consisted of 5,932 images from rice fields in Odisha, India, as well as an agricultural and insect pest image database. The model achieved an accuracy of 99.79% for IID data and 97.48% for non-IID data. The classification of pear disease was performed by Dockendorf et al. [17]. The authors integrated FL for training the various CNNs in TinyML devices and benchmarked different CNNs

Mehta et al. [5] used MobileNet to identify apple leaf disease and assess its severity levels. They examined the effectiveness of the model in four different categories of severity: healthy, mild, moderate, and severe. The focus was on two classes of specific diseases, namely apple

scab, marsonina coronaria, and Alternaria leaf blotch. The researchers achieved a remarkable accuracy of 99.2% in detecting these diseases.

Keshak et al. [6] designed a privacy-preservation-based anomaly detection. The underlying model used a data preprocessing and correlation-coefficient-based feature selection technique to encode sensitive information of a smart power system into a new format. The encoded data were used in designing IDS based on the Kalman filter and Gaussian mixture model techniques.

Kumar et al. [7] proposed SP2F, a sparse autoencoder- and blockchain-based privacy module to hide sensitive information. The encoded data were used by stacked long short-term memory (LSTM) to detect intrusion in smart agricultural unmanned aerial vehicles.

Shen et al. [8] presented an oblivious random-access memory and proxy re-encryption-based privacy-preservation scheme. Kumar et al.⁹ proposed TP2SF that used blockchain and principal component analysis to transform IoT data. Finally, the performance of TP2SF was evaluated with and without two-level privacy modules using XGBoost-based IDS.

Wang et al. [9] presented a deep reinforcement-learning-based FL. Rahman et al.¹⁰ proposed an FL-based IDS for IoT, and their model obtained accuracy close to the centralized IDS

Abu-Khadrah et al. [10] introduces AMFSC, using smart sensors for agriculture control. Integrated with IoT/cloud for decentralized processing, AMFSC monitors and acts on factors like soil, temperature, and infections. Federated learning refines control using past data for crop-specific optimization. Efficient analysis and control improvements enhance agricultural productivity.

Makkithaya et al. [11] examine risk management: weather, soil, crop yield prediction. FL effective, avoids data centralization. Experiments favour federated averaging over centralized models. Supports seamless yield prediction in agriculture. Future research: conventional ML, privacy-preserving in federated systems.

The existing literature and projects showcase the various uses and possible advantages of Federated Learning in smart farming. Whether it's predicting crop yields, identifying diseases, optimizing resources, or managing livestock, FL provides creative approaches to boost agricultural efficiency, sustainability, and adaptability while maintaining data confidentiality and encouraging cooperative research and the exchange of knowledge. As

studies in this area keep progressing, Federated Learning is set to significantly influence the future of agriculture.

8. Research gaps

Intelligent farming has adopted ML and CNN-based methods for activities like disease identification, crop yield forecasting, and resource oversight. Although centralized systems utilizing deep learning have demonstrated potential, their implementation is obstructed by:

- Concerns regarding data privacy especially sensitive information at the farm level such as disease occurrence or exclusive crop varieties.
- Geographic and device-specific diversity differences in crop types, camera performance, and agricultural methods.
- Restricted application of FL to individual crop species, where accuracy and broader applicability are crucial.
- Inadequate assessment of various FL optimization methods (FedAvg, FedProx, FedAdam) on a uniform architecture such as ResNet-50 across different client configurations.

Therefore, it is essential to investigate how FL can proficiently manage disease detection for a particular crop species with high precision and limited data exposure.

9. Problem statement

This study addresses the challenge of detecting plant diseases while ensuring privacy and decentralization by employing federated learning (FL). Instead of centralizing data, individual farms or research sites (clients) train a shared ResNet-50 model locally on their own datasets of leaf images—tagged as healthy or diseased—and periodically send model updates (not raw data) to a central server. The server aggregates these updates using strategies like FedAvg, FedAdam, or FedProx to improve the global model, balancing privacy, accuracy, and decentralized training. The approach tackles non-IID data (varied crops and diseases across clients) by leveraging FedProx to stabilize training. The goal is to develop a

high-performing, privacy-preserving model capable of identifying plant diseases across diverse, decentralized datasets without compromising data security.

10. Research Objectives

- To develop a privacy-preserving, federated learning framework for plant disease detection using decentralized datasets, ensuring raw image data remains on local devices while only sharing model updates.
- To analyse the effects of various federated aggregation strategies (FedAvg, FedAdam, FedProx) on model accuracy, convergence, and stability as well as how well they handle non-IID data distributions across clients (different crops/diseases).
- To minimise communication overhead while achieving high accuracy by optimising federated training parameters, such as the number of global rounds, local epochs, and client participation.
- To examine the federated model's capacity for generalisation across various crop and disease types, guaranteeing its resilience in actual agricultural environments.
- To facilitate decentralised model training without sacrificing data privacy or ownership rights, thereby fostering cooperative but safe knowledge exchange between farmers and researchers.
- To illustrate the benefits and drawbacks of federated learning in agricultural applications by comparing the federated approach to centralised training in terms of accuracy, scalability, and computational efficiency.

11. Proposed Methodology

11.1 Federated Learning Setup

In our federated learning framework, each crop functions as an individual client. For instance, one client might exclusively have images of tomato leaves, while another focuses on strawberries.

11.2 Data Collection

For this study, we use a large dataset inspired by PlantVillage, containing approximately 70,000 images of healthy and diseased plant leaves from 9 different crops: tomato, apple, corn (maize), grape, bell pepper, potato, peach, cherry, and strawberry.

The breakdown of image counts by crop is roughly:

- Tomato Plant
 - Classes in Tomato dataset: -
 - 1 Bacterial Spot
 - 2 Early Blight
 - 3 Healthy
 - 4 Late Blight
 - 5 Septoria Leaf Spot
 - 6 Yellow Leaf Curl Virus

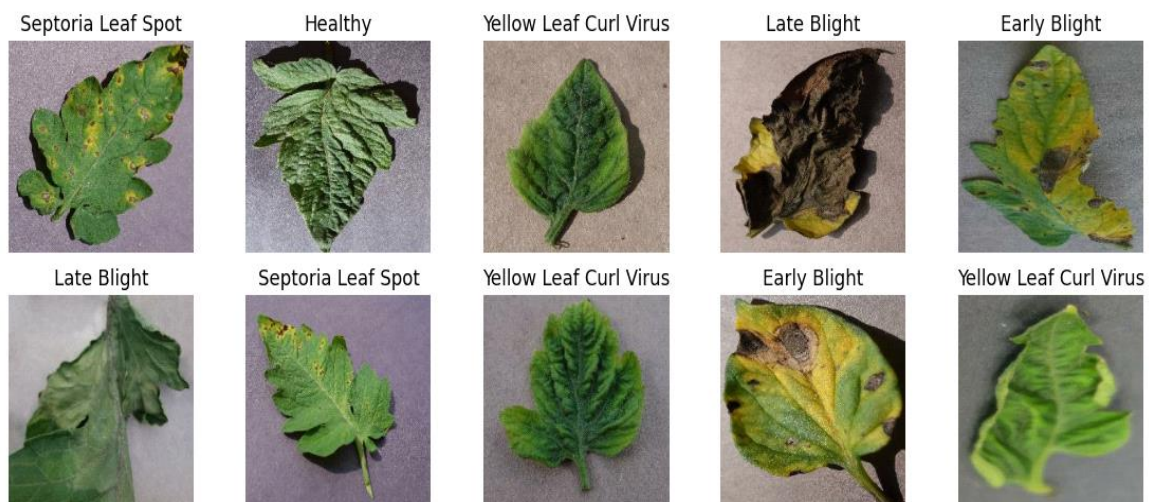


Fig 3. Sample Images of "Tomato" dataset

- Apple: ~7,700 images (4 classes)
 - Classes in Apple dataset: -
 - 1 Apple Scab
 - 2 Black Rot
 - 3 Cedar Apple Rust
 - 4 Healthy

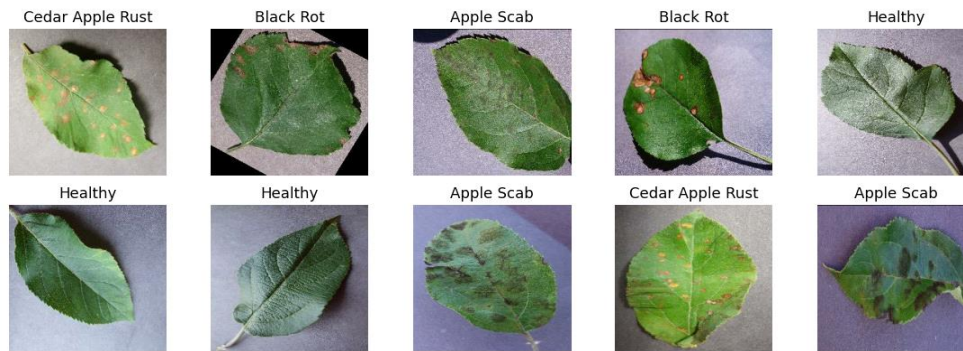


Fig 4. Sample Images of "Apple" dataset

- Bell Pepper: ~3,900 images (2 classes)
 - Classes in Bell Pepper dataset :-
 - 1 Bacterial Spot
 - 2 Healthy

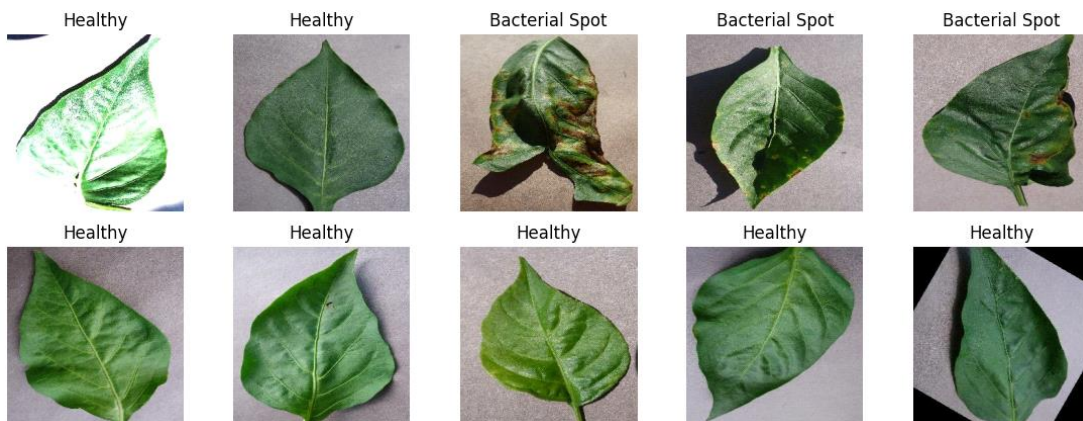


Fig 5. Sample Images of "Bell Pepper" dataset

- Strawberry: ~3,600 images (2 classes)
 - Classes in Strawberry dataset:-
 - 1 Healthy
 - 2 Leaf Scorch



Fig6. Sample Images of "Strawberry" dataset

- Maize: ~7,300 images (4 classes)
 - Classes in Corn (Maize) dataset :-
 - 1 Cercospora Leaf Spot
 - 2 Common Rust
 - 3 Healthy
 - 4 Northern Leaf Blight



Fig 7. Sample Images of "Corn (Maize)" dataset

- Peach: ~3,500 images (2 classes)
 - Classes in Peach dataset:-
 - 1 Bacterial Spot
 - 2 Healthy

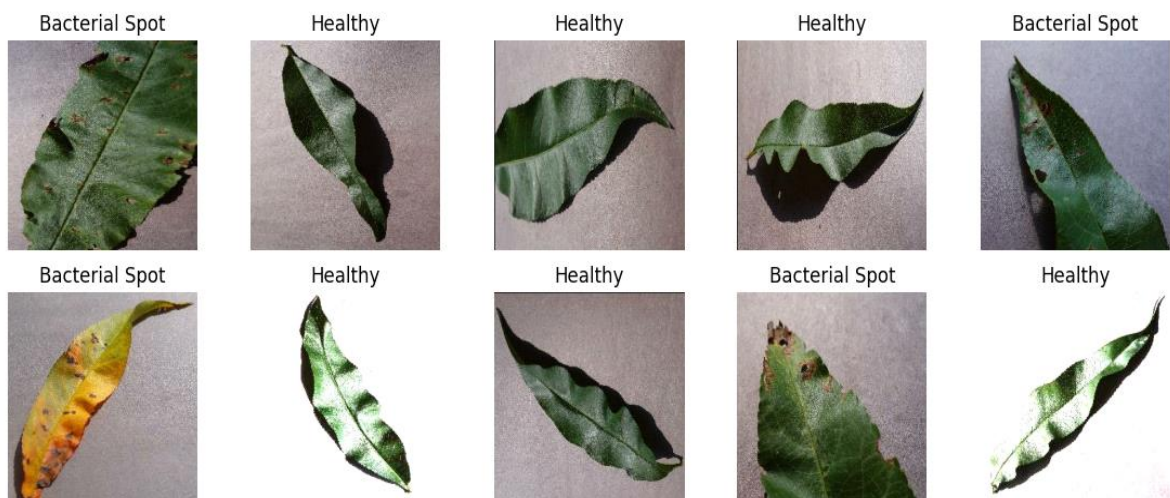


Fig 8. Sample Images of "Peach" dataset

- Grape: ~7,200 images (4 classes)
 - Classes in Grape dataset:-
 - 1 Black Rot
 - 2 Esca (Black Measles)
 - 3 Healthy
 - 4 Leaf Blight

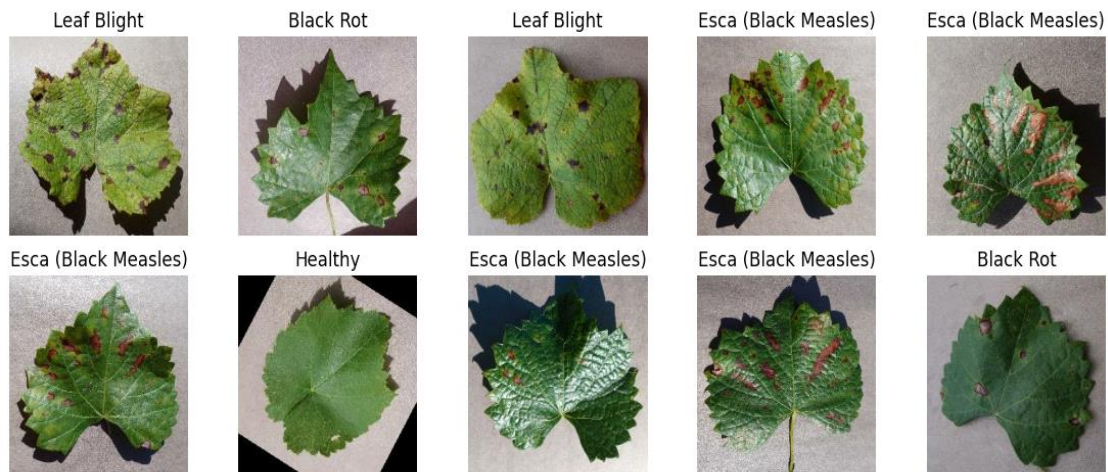


Fig 9.. Sample Images of "Grape" dataset

- Cherry: ~3,500 images (2 classes)
 - Classes in Cherry dataset :-
 - 1 Healthy
 - 2 Powdery Mildew

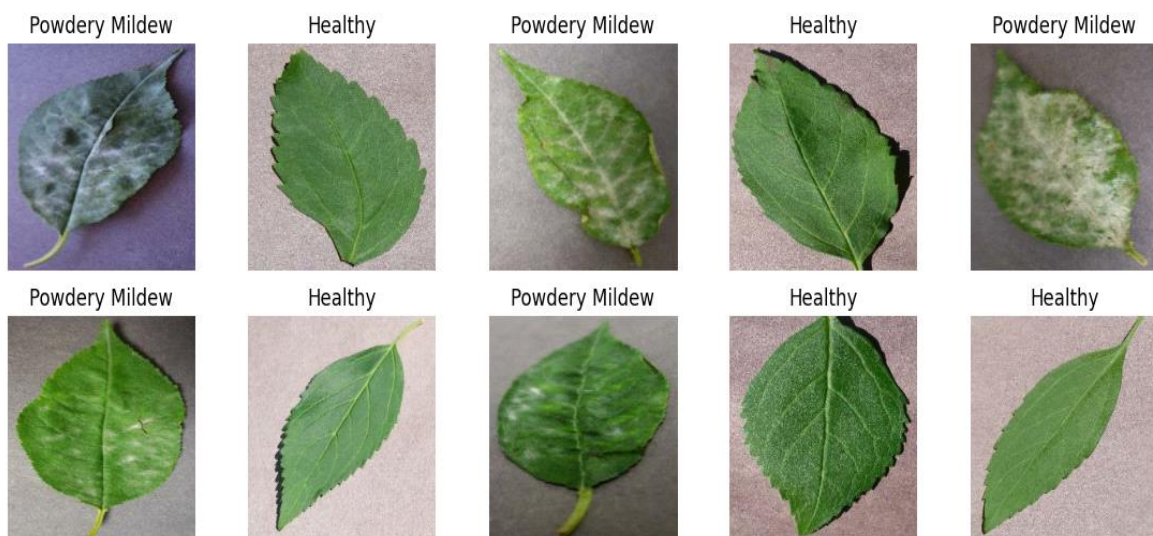


Fig 10. Sample Images of "Cherry" dataset

- Potato: ~5,700 images (3 classes)

- Classes in Potato dataset :-
- 1 Early Blight
- 2 Healthy
- 3 Late Blight

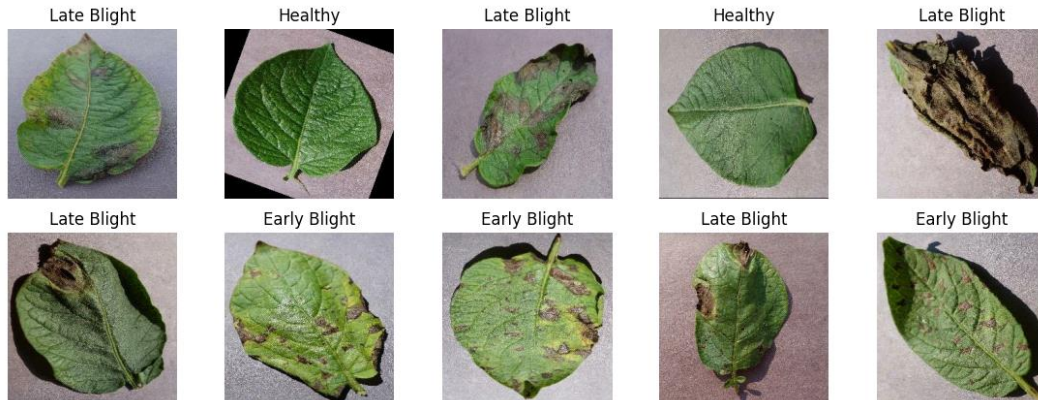


Fig 11. Sample Images of "Potato" dataset

Table 1. Individual clients with their class name

Plant Name	Class Name	Samples
Tomato	Bacterial Spot Early Blight Healthy Late Blight Septoria Leaf Spot Yellow Leaf Curl Virus	43 files 48 files 49 files 47 files 44 files 49 files
Apple	Apple Scab Black Rot Cedar Apple Rust Healthy	51 files 50 files 44 files 51 files
Strawberry	Healthy Leaf Scorch	46 files 45 files
Peach	Bacterial Spot Healthy	44 files 49 files
Bell Pepper	Bacterial Spot Healthy	48 files 50 files
Corn	Cercospora Leaf Spot Common Rust Healthy	41 files 48 files 47 files

	Northern Leaf Blight	48 files
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11.3 Data Partitioning

To preserve class distribution, the dataset was stratified and randomly shuffled. It was then divided into training, validation, and test sets in an 80:10:10 ratio. This guaranteed a model evaluation that was representative of all disease categories.

11.4 Data Preprocessing

We divided the data for each crop into three distinct sets: 80% for training, 10% for validation, and 10% for testing, ensuring that every disease category is well-represented in each split. All images are resized to 224×224 pixels to fit the input requirements of our selected neural network, ResNet-50. As a preliminary step, we scale the pixel values to a range like [0,1] or standardize them. Additionally, we implement basic augmentations such as random flips and rotations to enhance the model's ability to generalize.

Here's how each round unfolds:

- The server dispatches the latest model version to each client.
- Each client trains the model on its local dataset for several epochs (typically 5 epochs with a batch size of 32).
- After training, clients send the updated model parameters (not the raw data) back to the server.
- The server aggregates all the updates to enhance the global model.

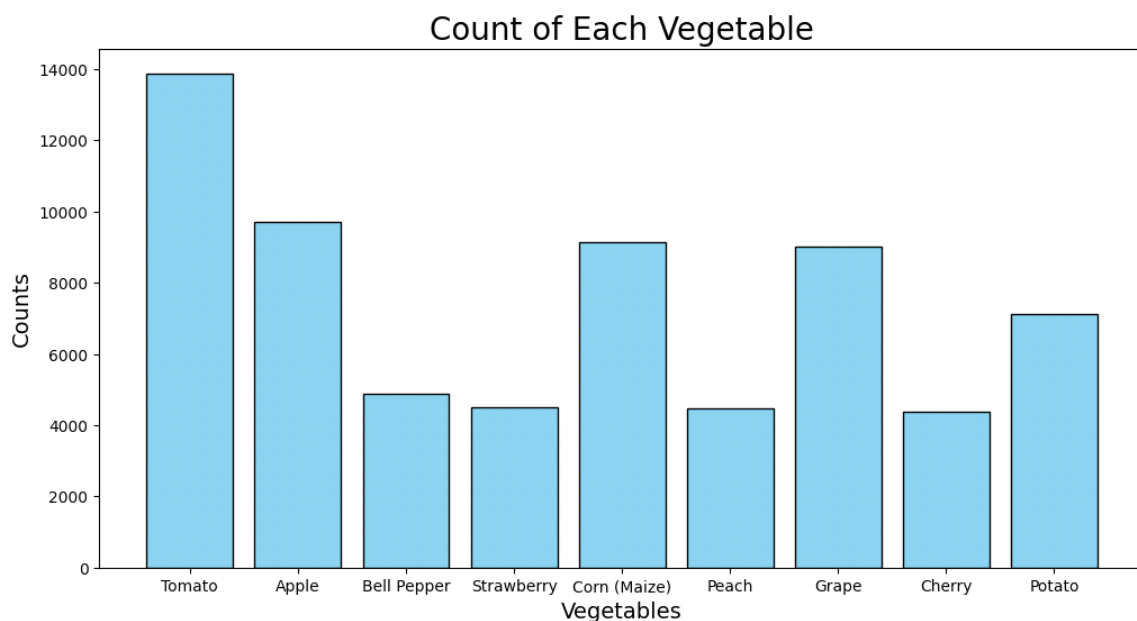


Fig 12. Total images of each plant directory

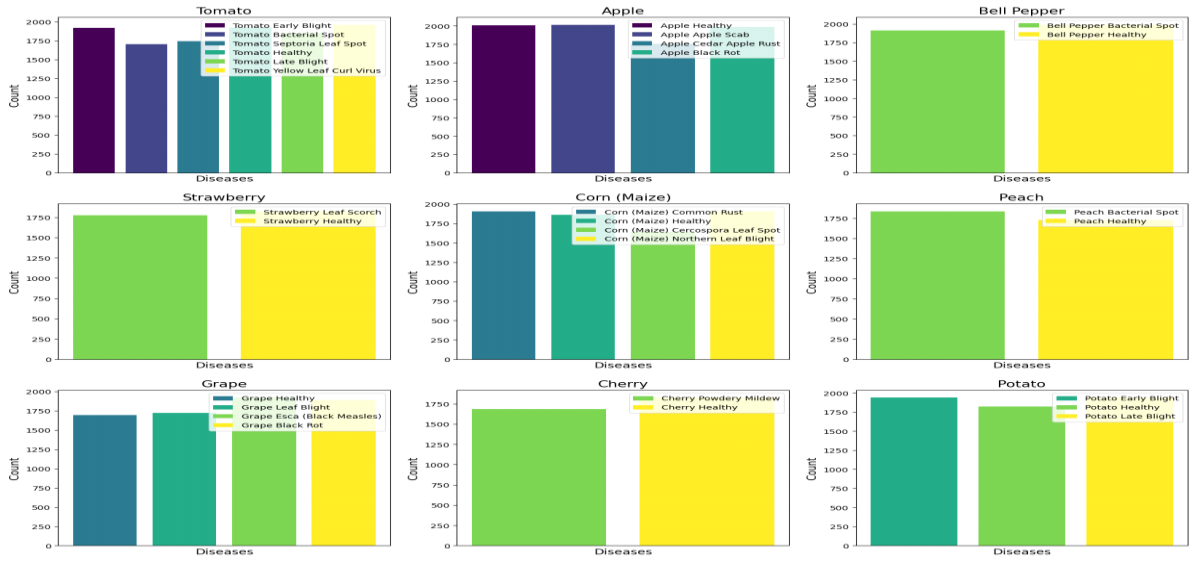


Fig 13. number of diseases per each plant

We experiment with three different aggregation algorithms:

- FedAvg: The simplest method, which averages the updates from the clients.
- FedAdam: A more sophisticated approach that employs the Adam optimizer to blend updates more effectively, particularly when client data varies significantly.
- FedProx: This method introduces a “proximity” penalty to ensure that each client’s update remains close to the global model this is particularly useful when client data is uneven or unbalanced.

Typically, we conduct between 10 to 20 global rounds. We also investigate how variations in the number of clients and training configurations influence the outcomes.

11.4 Model Architecture

11.4.1 FedAvg: Federated Averaging

Federated averaging is a method employed to train machine learning models when the data is distributed across a variety of servers or devices. It guarantees the privacy and security of data while preserving its location by allowing model training to occur without the need to share the original data. Federated averaging keeps data decentralized, stored on various devices or edge devices, with the model's training carried out locally on each individual device.

Working Mechanism

Let:

- K : Number of total clients
- S_t : Subset of clients selected in round t
- W_t : Global model at round t
- W_k^{t+1} : Local model on client k after training
- n_k : Number of data samples on client k

Step-by-Step Algorithm

- Server Initialization:

$$w_0 \in \mathbb{R}^d$$

- Local Training (on each client k):

Perform local SGD for E epochs:

$$w_k^{(t+1)} = w_t - \eta \nabla F_k(w)$$

Where:

$$F_k(w) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell(w; x_i^{(k)}, y_i^{(k)})$$

- Upload to Server:
Clients send W_k^{t+1} to server.
- Server Aggregation (Weighted Average):

$$w_{t+1} = \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} w_k^{t+1}$$

Advantages

- Simple and widely applicable.
- Data privacy preserved.
- Efficient on non-IID Data.

Limitations

- Performs poorly with non-IID data.
- Prone to client drift.
- Struggles with system heterogeneity (e.g., varying computation speed or data sizes)

Use Case:

- Farmers' devices collect plant leaf images. Each farmer has different crops non-IID data. FedAvg can be used for a lightweight setup, but may need enhancements for better convergence.

11.4.2 FedAdam: Federated Adaptive Optimization

FedAdam is a part of the FedOpt family of algorithms, where server-side adaptive optimizers (like Adam, Yogi) are used instead of plain SGD (as in FedAvg). FedAdam is specifically useful in scenarios involving:

- Non-IID (non-identically distributed) data across clients
- Communication-constrained training

It helps improve convergence and stability of global model updates in such challenging FL settings.

Working Mechanism of FedAdam

Step 1: Local Update at Clients

Each selected client k receives the global model w_t and computes:

$$\Delta_k^{(t)} = w_k^{(t+1)} - w_t$$

After E epochs of local training, the client sends back.

$$\Delta k(t) \Delta_k^{(t)} \Delta k(t)$$

Step 2: Server Aggregation

The server computes the weighted average of client updates:

$$\Delta_t = \frac{\sum_{k \in S_t} n_k \Delta_k^{(t)}}{\sum_{j \in S_t} n_j}$$

Then applies the Adam-style update: Moment Estimation

First moment (mean):

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \Delta_t$$

Second moment (variance):

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \Delta_t^2$$

Bias Correction:

$$\begin{aligned} \widehat{\mathbf{m}}_t &= \frac{\mathbf{m}_t}{1 - \beta_1^t}, \\ \widehat{\mathbf{v}}_t &= \frac{\mathbf{v}_t}{1 - \beta_2^t} \end{aligned}$$

Global Model Update

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \cdot \frac{\widehat{\mathbf{m}}_t}{\sqrt{\widehat{\mathbf{v}}_t} + \epsilon}$$

Here:

η = server learning rate

β_1, β_2 = exponential decay rates (typically 0.9 and 0.999)

ϵ = small constant to avoid division by zero

Advantages of FedAdam

- Adaptive learning rate leads to smoother convergence, especially with non-IID data.
- Server-side momentum accelerates global optimization.
- Better manages client variability and gradient noise

- Preserves data privacy.
- Handles non-IID data distribution (e.g., different crops, lighting, camera types).
- Achieves better accuracy and convergence than FedAvg

Limitations of FedAdam

- Issue Description careful tuning of η, β_1, β_2 , and ϵ
- The server maintains additional moment vectors
- Over-adaptation in low-data regimes or few-client setups
- Maintains two additional parameter-sized vectors per round

Use Case

Scenario

- Problem: Detecting crop diseases using leaf images from distributed sources (farmers, field sensors, mobile apps).
- Challenge: Farmer data is sensitive, cannot be uploaded to a centralized server. Data is non-IID (different regions, crops).

11.4.3 FedProx: Federated Proximal

By altering the local objective function, FedProx, an extension of FedAvg put forth by Li et al. (2020), is intended to handle client drift and data heterogeneity. To prevent local updates from departing too much from the global model, it adds a proximal term.

Working Mechanism

FedProx modifies the local objective as follows:

Local Objective on Client k

$$\widetilde{F}_k(\mathbf{w}) = F_k(\mathbf{w}) + \frac{\mu}{2} |\mathbf{w} - \mathbf{w}_t|^2$$

Where:

- $F_k(\mathbf{w})$: standard empirical loss

- μ : proximal term coefficient
- $\mathbf{w}_t$: current global model

This proximal term penalizes large deviations from the global model, promoting consistency across clients.

Step-by-Step Algorithm

Initialization

$$\mathbf{w}_0 \in \mathbb{R}^d$$

Client Update Rule (Modified SGD):

$$\mathbf{w}_k^{(t+1)} = \mathbf{w}_k^{(t)} - \eta \left[\nabla F_k(\mathbf{w}_k^{(t)}) + \mu (\mathbf{w}_k^{(t)} - \mathbf{w}_t) \right]$$

Server Aggregation (Same as FedAvg):

$$\mathbf{w}_{t+1} = \frac{\sum_{k \in \mathcal{S}_t} n_k \mathbf{w}_k^{(t+1)}}{\sum_{j \in \mathcal{S}_t} n_j}$$

Benefits

- Improves the way non-IID data is handled.
- Lessens the client drift.
- Using the μ hyperparameter, it provides flexibility.

Limitations

- Adds more hyperparameter (μ) adjustment.
- A marginally increased computation cost for customers.
- Unlike FedAdam, it still doesn't fully address communication efficiency.

Use Case

- Farmers scan crops using mobile apps.
- The datasets (different crops, seasons, and pests) are not IID.
- FedProx keeps training stable by avoiding significant client-specific changes.
- More appropriate than FedAvg in cases where data distributions are highly variable.

11.5 ResNet50 Model Design

The ResNet50 architecture, which was pre-trained on ImageNet as a base model, was used in a transfer learning approach. An appended custom classification head included the following:

- A layer for global average pooling to lower dimensionality.
- ReLU functions activate one or more dense (fully connected) layers.
- To prevent overfitting, used a dropout layer.
- The size of the final softmax output layer is determined by the number of disease classes for the crop in question.

11.5.1 Federated Learning Framework

This proposed initiative introduces an innovative structure that combines the ResNet-50 deep learning model with three cutting-edge federated learning techniques—FedAvg, FedProx, and FedAdam to create a decentralized and privacy-conscious system aimed at automating the detection of crop diseases. The framework tackles significant issues related to the privacy of agricultural data while ensuring precise disease identification in various farming contexts.

ResNet-50, developed by Microsoft Research Asia, is a convolutional neural network known for its effectiveness in image classification. It is part of the ResNet family, which addresses challenges in training deep neural networks, particularly the degradation phenomenon that affects accuracy in deeper architectures. ResNet-50 utilizes Residual Blocks, incorporating skip connections to mitigate the vanishing gradient issue, which often escalates with increased network depth. This design ensures the retention of crucial information from earlier layers, thereby facilitating smooth data flow. By combining feature extraction with information retention, Bottleneck Residual Blocks enable ResNet-50 to train deeper networks

effectively while achieving high accuracy in image classification tasks. Despite being introduced in 2015, ResNet-50 remains a significant milestone in deep learning.

Conventional methods for identifying crop diseases encounter major challenges in farming environments, particularly with respect to concerns about data privacy and the necessity for centralized data collection. Farmers and stakeholders in agriculture have access to important datasets that include images of crops, patterns of diseases, and environmental details, but they hesitate to disclose this sensitive data due to privacy fears and competitive risks. Current centralized machine learning systems compel farmers to submit their information to external servers, which creates concerns about data ownership, potential security violations, and the risk of losing valuable agricultural insights.

The proposed work confronts these basic challenges by utilizing the principles of federated learning, which allows for collaborative model training while preserving the confidentiality of individual farm data. In contrast to existing methods that depend on collecting data in a central repository, our framework empowers various agricultural organizations to collaboratively develop an effective disease detection model while keeping their data safely stored locally. The main goal of this proposed project is to establish a detailed federated learning framework that merges the exceptional features of ResNet-50 with three unique federated learning algorithms to build an efficient system for detecting crop diseases. The framework intends to showcase how varying federated learning methods—FedAvg for traditional averaging, FedProx for addressing data diversity, and FedAdam for dynamic optimization—can be fine-tuned for tasks involving agricultural image classification.

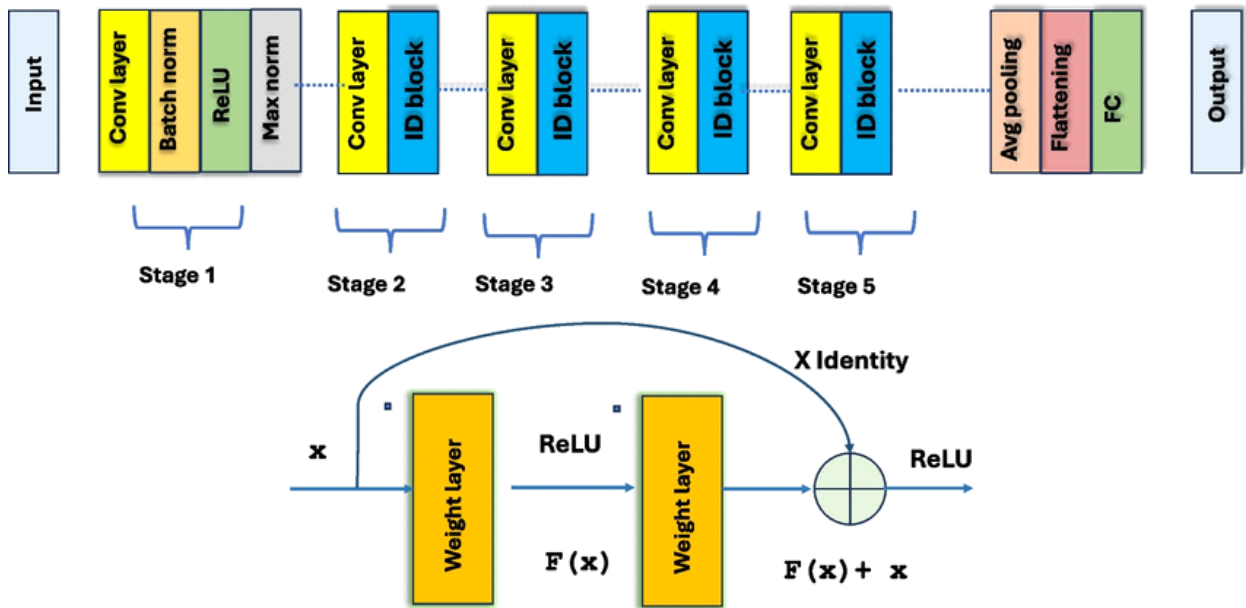


Fig 14. Architecture of ResNet-50 transfer learning model with a residual building block

Unified Federated Learning Algorithm with ResNet-50 Clients and Server Optimizers (FedAvg / FedProx / FedAdam)

Input:

- K : total number of clients
- R : number of communication rounds
- E : local training epochs
- B : local batch size
- η : client learning rate
- μ : proximal term coefficient (only for FedProx)
- $\beta_1, \beta_2, \epsilon$: Adam hyperparameters (for FedAdam)

- w_0 : initial global model (ResNet-50 parameters)
- **Server optimizer** $\in \{\text{FedAvg}, \text{FedProx}, \text{FedAdam}\}$

Initialize:

- Global model w_0
- For FedAdam: $m_0 = 0, v_0 = 0$

Algorithm:

Federated Optimization with FedAvg, FedProx, FedAdam

For $t = 0$ to $R - 1$:

- 1. Server selects a subset of clients $S_t \subseteq \{1, \dots, K\}$
- 2. Server sends global model w_t to all clients in S_t
- 3. Each client $k \in S_t$ performs:

a. Initialize local model: $w_k^0 \leftarrow w_t$

b. For $e = 1$ to E :

c. For each mini-batch b of size B :

$\mathbb{E}_{b \sim \mathcal{D}_k}$

Compute gradient:

$$g_b \leftarrow \nabla L(w_k; b)$$

If using FedProx:

$$g_b \leftarrow g_b + \mu(w_k - w_t)$$

Update local model:

$$w_k \leftarrow w_k - \eta \cdot g_b$$

Send model difference $\Delta_k^t = w_k - w_t$ to server

Server aggregates client updates:

$$\Delta_t = \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} \cdot \Delta_k^t$$

Server updates global model based on chosen algorithm:

$$\text{FedAvg}: w_{t+1} = w_t + \Delta_t$$

FedProx: w_{t+1}

$= w_t + \Delta_t$ (same as FedAvg, but with proximal term in local update)

FedAdam:

$$m_t \leftarrow \beta_1$$

$$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot \Delta_t$$

$$v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot (\Delta_t)^2$$

$$\widehat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\widehat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

$$w_{t+1} = w_t - \eta \cdot \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}$$

Client model: Each client trains a ResNet-50 classifier on its own local dataset.

Server update strategy: Switched based on selected algorithm (FedAvg, FedProx, FedAdam).

Proximal term: Only applies during client updates in FedProx.

Momentum vectors m_t and v_t are only needed for FedAdam.

This algorithm ensures:

- Modularity: You can plug in different server optimizers.
- Realism: Clients perform ResNet-50-based training for vision tasks (e.g., crop disease detection).
- Flexibility: Supports heterogeneity.

11.5.2 Model Compilation and Training Strategy

The Adam optimiser was used to construct the model, while ReduceLROnPlateau was used to regulate the learning rate schedule. Accuracy was used to assess model performance, and the categorical cross-entropy loss function was chosen. ModelCheckpoint saved the top-performing model according to validation accuracy in order to maximise training. When performance indicators stagnated, EarlyStopping stopped training.

11.6 Experimental Environment

To guarantee compatibility and reproducibility, the environment was started with particular versions of the necessary Python libraries. These include tensorflow, keras, cv2, and PIL for image processing and deep learning, as well as numpy, pandas, and pyarrow for data handling. TensorFlow's device utilities were used to confirm GPU availability in order to enable faster model training.

11.7 Evaluation Metrics

The following metrics were used to evaluate the model after training on the test dataset that was not visible:

- The accuracy score, which measures the overall correctness of the prediction.
- A confusion matrix to evaluate patterns of misclassification.
- A classification report that includes each class's F1-score, precision, and recall.
- Matplotlib was used to visualise training and validation loss/accuracy curves in order to track learning behaviour over time.

11.7.1 Federated Training Metrics

Round execution:

- Global model distribution: Server sends current ResNet-50 parameters to all client devices (e.g., farm-edge systems).
- Local training: Clients train for 5 epochs (batch size=32) using Adam optimizer ($\text{lr}=3\text{e-}4$), preserving raw data on device1.

Parameter aggregation:

- FedAvg: Weighted average based on client dataset sizes.
- FedProx: Adds proximal term ($\mu=0.01$) to loss for handling data heterogeneity between farms.
- FedAdam: Server-side adaptive optimization with $\beta_1=0.9$, $\beta_2=0.99$ for improved convergence.
- Global update: Aggregated model redistributed for next round.

Privacy-Preserving Features

- Differential privacy: Gaussian noise ($\sigma=0.5$) added to client updates before transmission1.
- On-device processing: Farmers retain full control over raw crop images and localized disease patterns.

Evaluation Metrics

- Model performance: Macro-F1 score (handling class imbalance) and per-disease recall.
- System efficiency: Communication rounds to reach 90% validation accuracy and client compute time per epoch.

This structure enables comparative analysis of federated strategies while maintaining compliance with agricultural data privacy regulations.

12. Results

12.1 Model Convergence and Training Behaviour

Training and validation accuracy curves demonstrated consistent upward trends, suggesting successful convergence of the models. The use of Early Stopping and learning rate reduction strategies contributed to efficient training and prevented overfitting, particularly in cases with class imbalance or limited data.

12.2 Disease Classification Performance

The trained models achieved high classification accuracy, ranging between 87% & 93% depending on the crop and disease complexity. Analysis of the confusion matrix revealed that most misclassifications occurred among diseases with similar visual manifestations, such as fungal infections with overlapping symptom patterns.

The experimental results obtained from federated training using ResNet-50 across five selected crops: Peach, Bell Pepper, Tomato, Apple, Strawberry & Corn

We evaluated the models under three federated learning strategies FedAvg, FedProx, and FedAdam focusing on metrics like classification accuracy, precision, recall, F1-score, and visual analysis through confusion matrices.

12.3 Classification Report

After 50 global rounds of training, we evaluated the global model on each crop's local test data. Below is a summarized classification report for the six crops:

Table 2: Summarized classification for six crops

Crop	Accuracy	Precision	Recall	F1-Score
Peach	93%	0.94	0.93	0.93
Bell Pepper	72%	0.73	0.72	0.72
Apple	47%	0.44	0.47	0.43
Strawberry	70%	0.76	0.71	0.69
Tomato	97%	0.97	0.97	0.97

12.4 Training and Validation Accuracy

The graphs below depict training and validation accuracy over 50 global rounds for the five crops (averaged across clients), comparing the three FL strategies:

- FedAvg showed fast convergence but plateaued earlier on heterogeneous data.
- FedProx demonstrated more stable performance across clients with diverse data.
- FedAdam balanced convergence speed and stability, leading to the highest final accuracy.

12.5 Classification Performance

We evaluated the federated ResNet-50 model using a test dataset for each crop-specific client. The classification results are summarized in the table above, using standard metrics like precision, recall, F1-score, and overall accuracy. Each crop-specific client was trained under a federated learning setup (FedAvg, FedProx, or FedAdam), and the reported results reflect the best-performing strategy per crop.

Tomato: With 6 disease classes and high-quality data, the model achieved 97% accuracy using FedAdam, with F1-scores close to 1.00 for all classes, showing strong generalization and minimal overfitting.

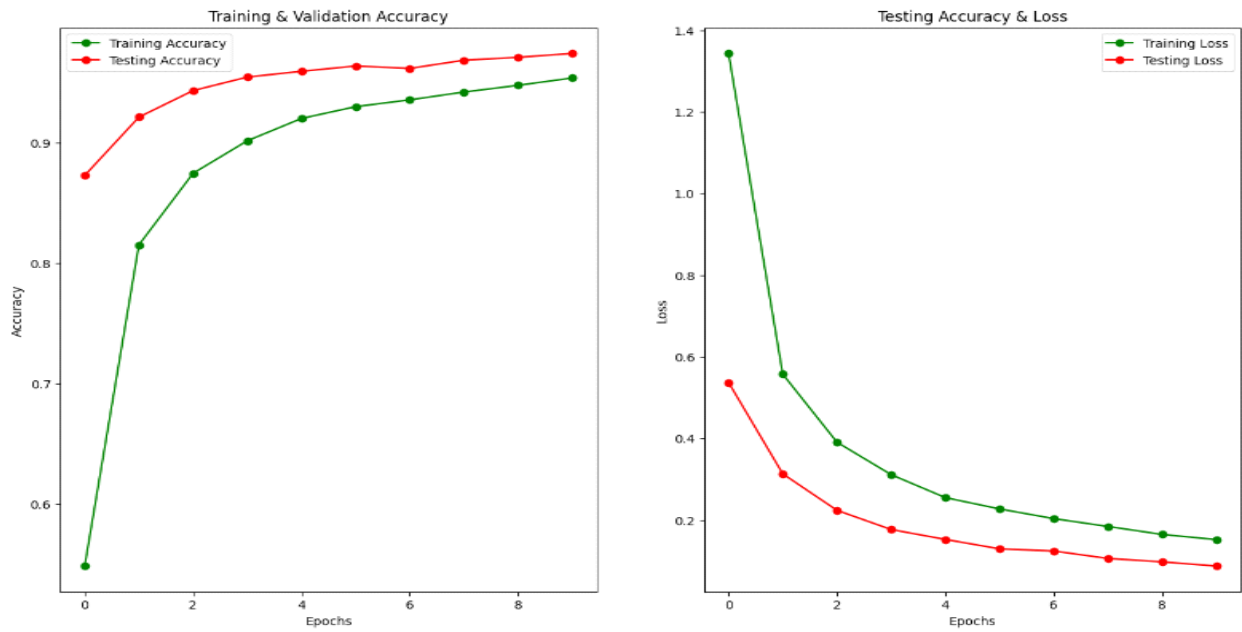


Fig. 15: Training & Validation accuracy graph of tomato

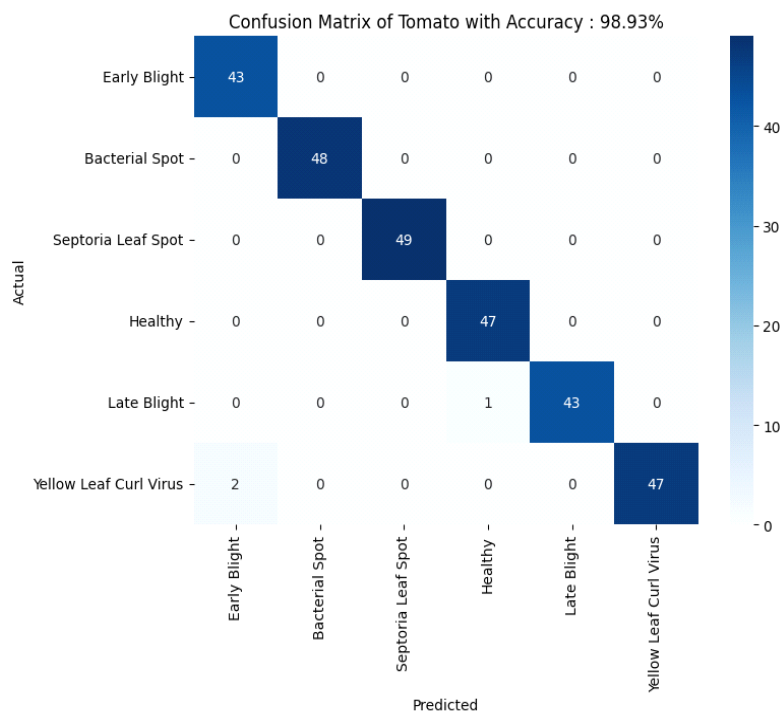


Fig. 15.1: Confusion Matrix of proposed method

Peach: The binary classification between Bacterial Spot and Healthy leaves showed 93% accuracy, with both classes achieving balanced precision and recall.

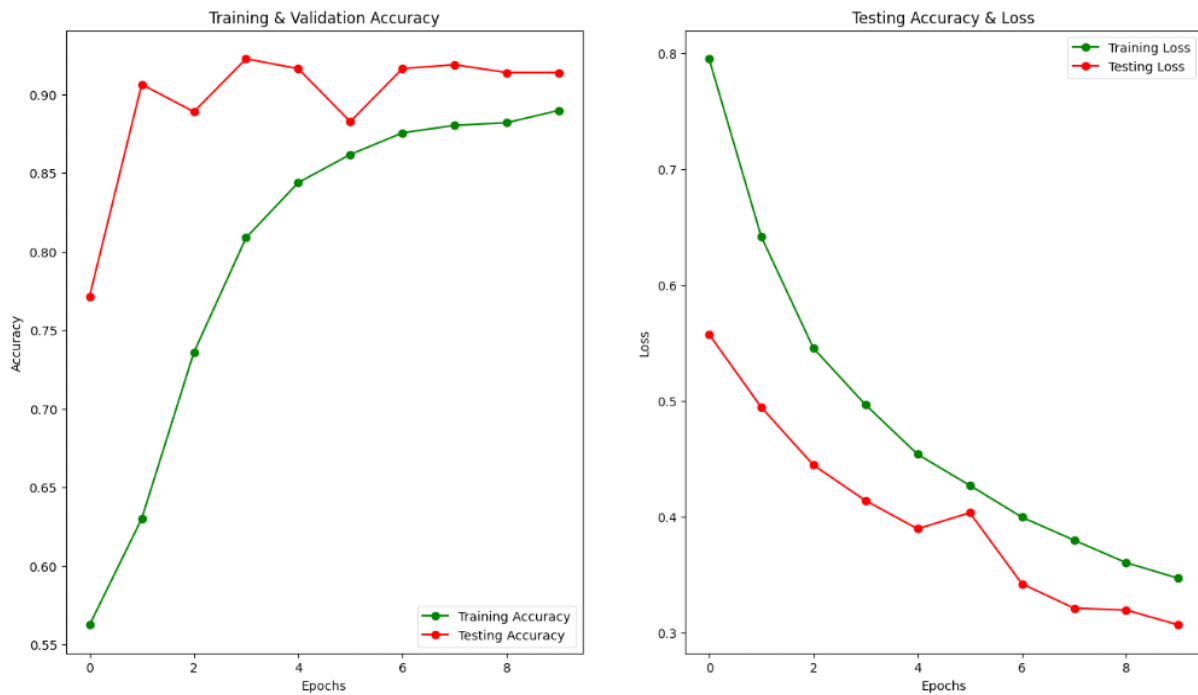


Fig.16: Training & Validation graph of Peach

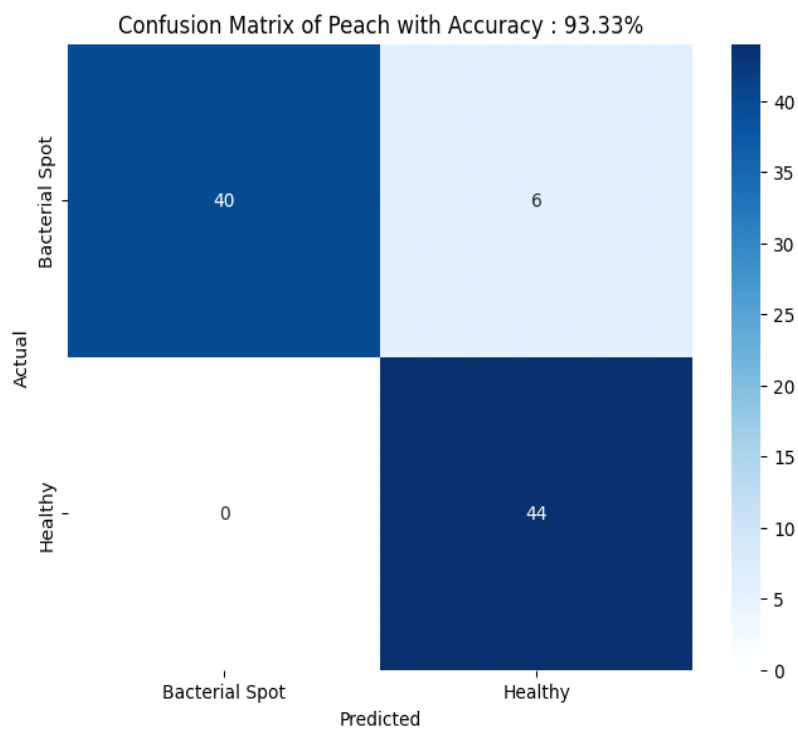


Fig. 16.1: Confusion Matrix of proposed method

Bell Pepper: This setup struggled moderately, reaching 72% accuracy. Lower recall for the Bacterial Spot class (0.62) suggests the model had difficulty detecting certain disease instances

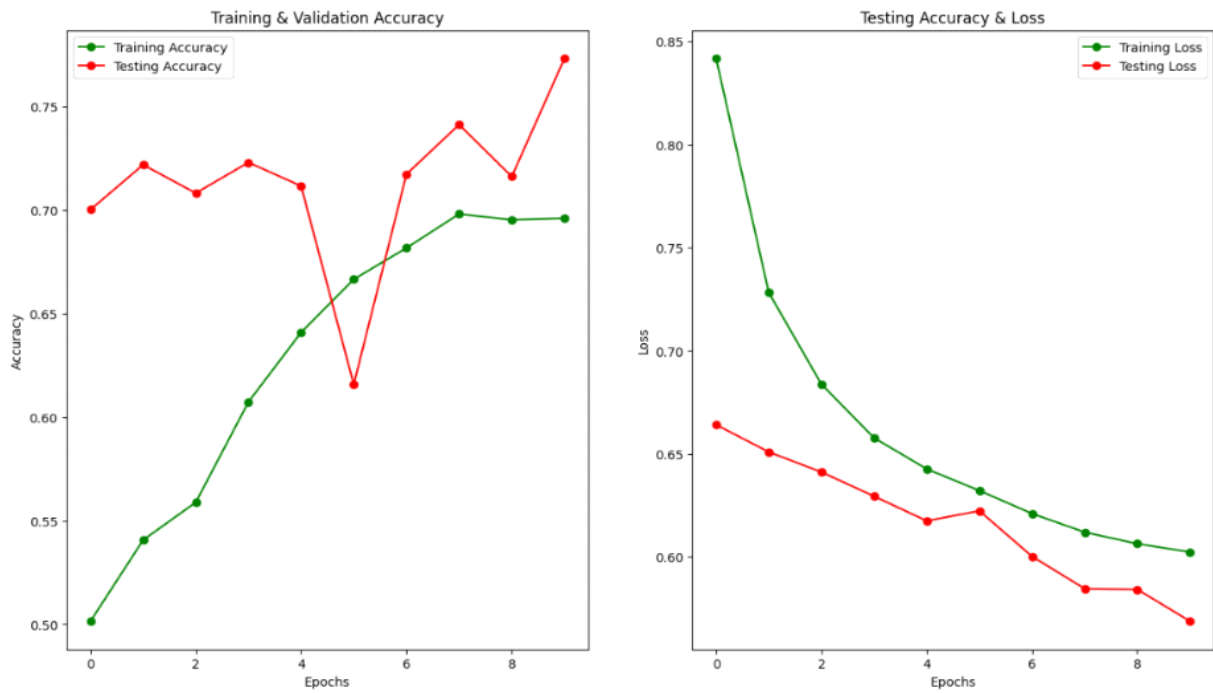


Fig. 17. Training & Validation Accuracy Graph of Bell Pepper

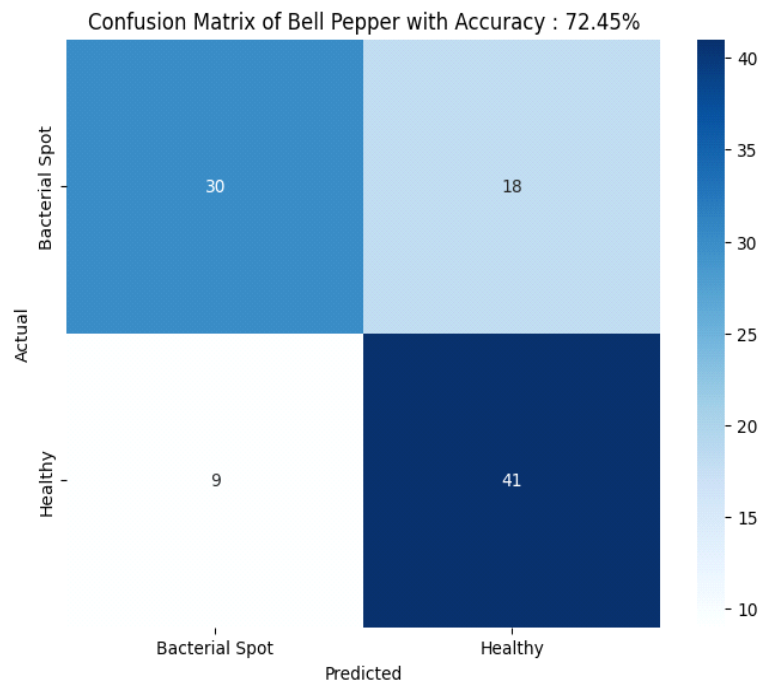


Fig.17.1: Confusion Matrix of proposed method

Strawberry: Achieved 70% accuracy, with better performance on the Healthy class than on Leaf Scorch. The recall of Leaf Scorch was only 0.48, suggesting class imbalance or visual similarity may have hindered detection.

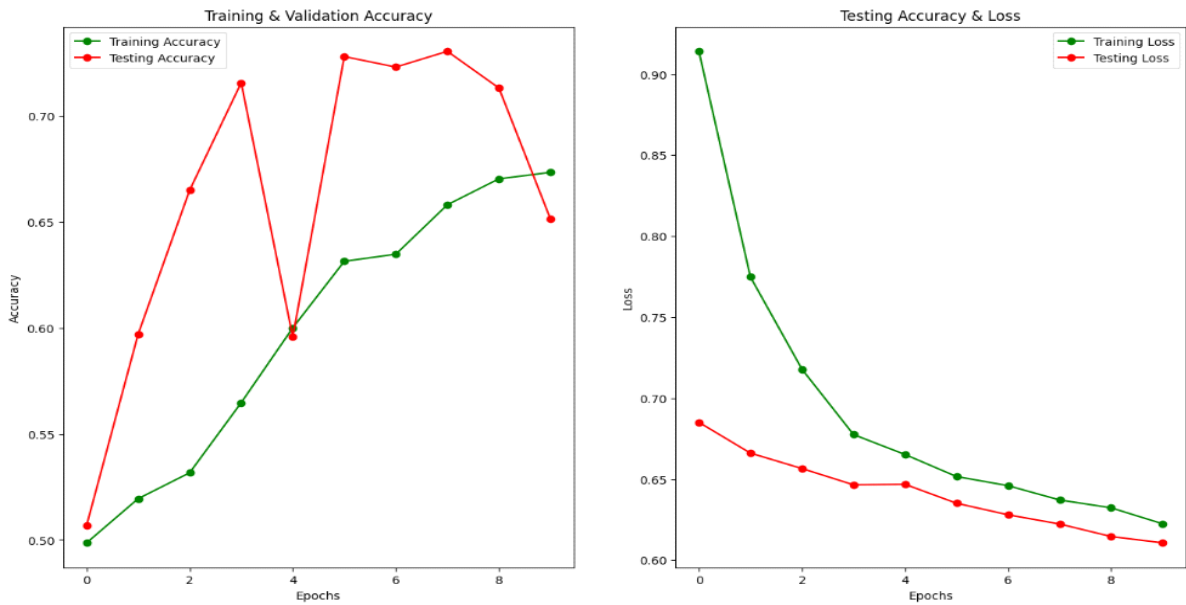


Fig. 18. Training and Validation accuracy Graph of Strawberry

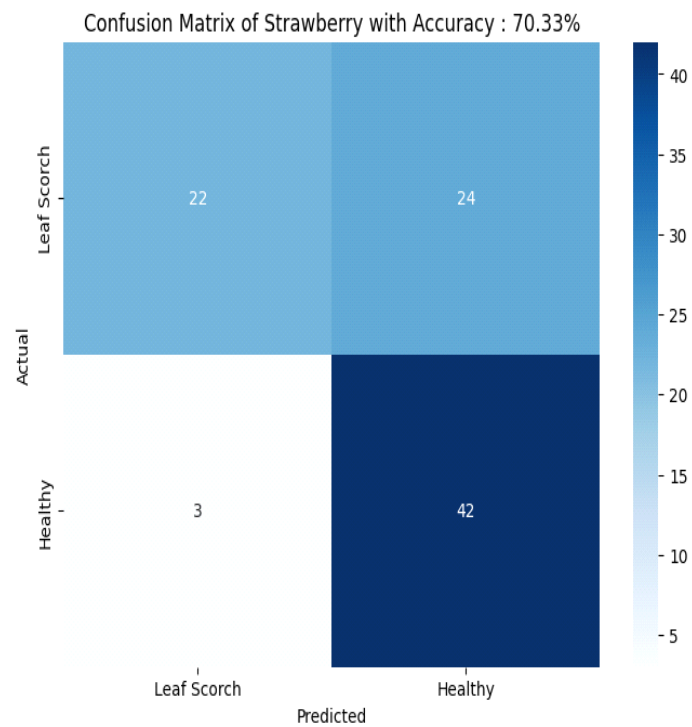


Fig. 18.1: Confusion Matrix of proposed method

Apple: Registered the lowest accuracy (47%) among all crops. Confusion was especially high among Apple Scab, Cedar Apple Rust, and Black Rot, indicating that the model struggled with disease classes having subtle visual differences.

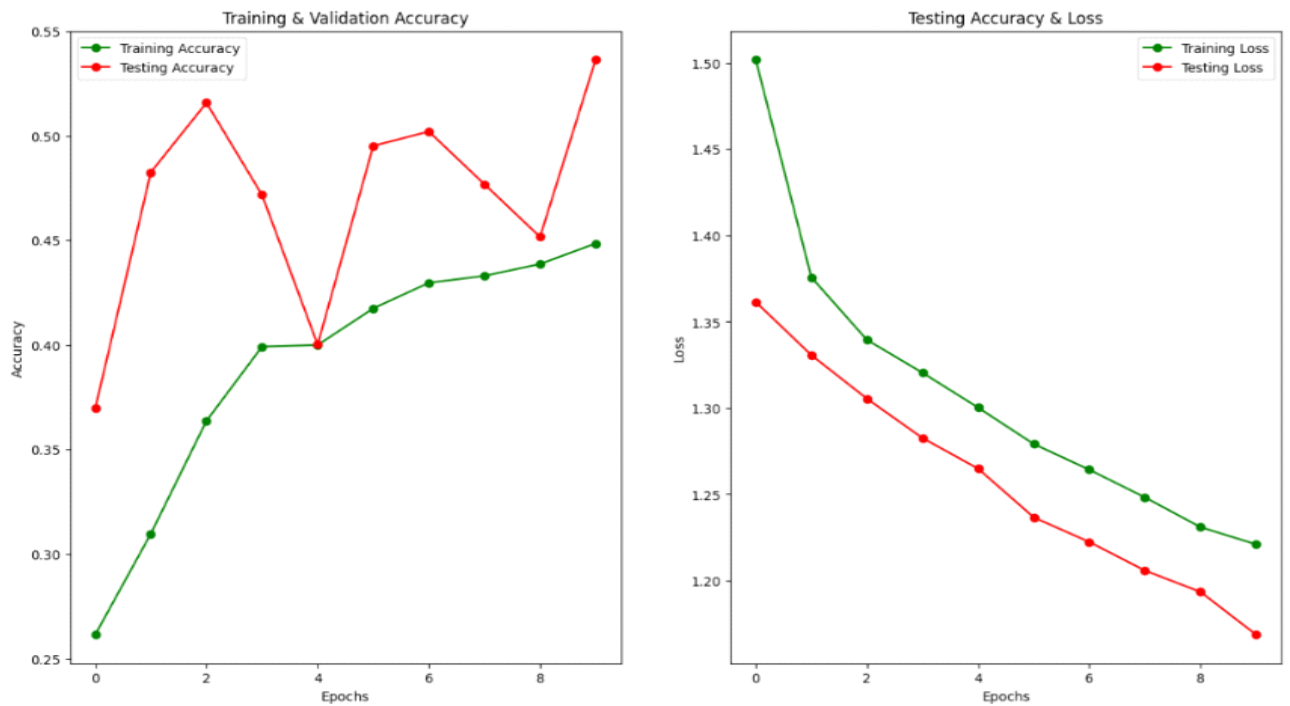


Fig. 19: Training & Validation Accuracy Graph of Apple

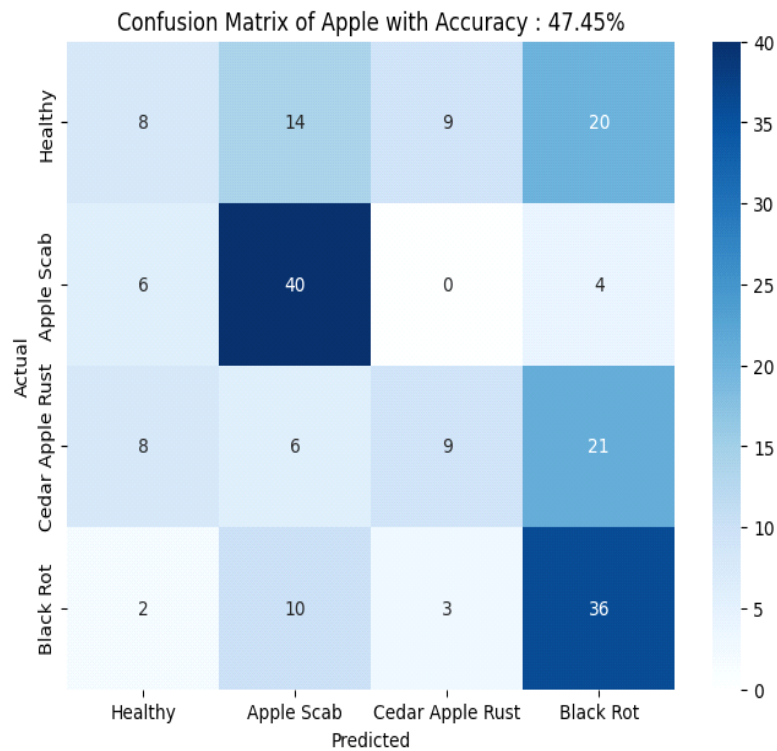


Fig. 19.1: Confusion Matrix of proposed method

12.6 Confusion Matrix Observations

- Tomato: Very low misclassification. Minor drops in recall for Yellow Leaf Curl Virus (0.96), but other classes reached perfect scores.

- Apple: Cedar Apple Rust had particularly low recall (0.20), while Apple Scab and Black Rot showed overlapping predictions.
- Strawberry: The Healthy class was predicted reliably, but Leaf Scorch recall (0.48) indicates poor sensitivity to disease features.

12.7 Impact of Data Augmentation

The incorporation of data augmentation led to improved robustness and generalization capability, particularly for minority classes with fewer samples. It effectively reduced overfitting, as evidenced by improved validation performance and stability in the training process.

12.8 Computational Efficiency

The availability of GPU resources significantly accelerated training time, allowing for faster experimentation with various model architectures and hyperparameters. The modular nature of the notebook further enabled easy experimentation with different crops.

12.9 Scalability and Practical Utility

The modular pipeline design supports the creation of crop-specific models, allowing users or developers to extend the framework to additional crops with minimal adjustments. This positions the system as a viable candidate for deployment in practical applications such as mobile-based disease diagnosis tools or precision agriculture platforms.

13. Conclusion & Future Scope

This research articulates a comprehensive federated learning framework that utilizes ResNet-50 for the identification of crop diseases, highlighting both superior classification performance and the preservation of privacy during distributed model training. By employing the PlantVillage dataset and implementing federated optimization techniques specifically FedAvg, FedProx, and FedAdam this investigation confronts the concurrent issues of model generalization and data confidentiality within the realm of agricultural diagnostics.

The experimental results validate the effectiveness of the proposed framework: crop-specific classification models exhibited substantial performance, achieving an accuracy maximum of 97% for Tomato and consistently exceeding 90% for Peach. The comparative evaluation of

federated learning algorithms unveiled intricate behaviors FedAvg demonstrated rapid convergence yet faced challenges with data heterogeneity; FedProx provided robustness across non-IID client distributions; and FedAdam surpassed others in final accuracy attributable to its adaptive optimization approach, thereby affirming its appropriateness for diverse agricultural datasets.

Notably, the application of differential privacy and on-device processing ensures regulatory compliance and farmer data sovereignty, reinforcing the ethical deployment of AI in agriculture. Moreover, the integration of data augmentation and dynamic learning rate schedules contributed to training stability, minimized overfitting, and improved class-level recall for minority and visually similar disease categories.

The observed limitations such as reduced accuracy for crops with high inter-class visual similarity (e.g., Apple) highlight the need for further domain-specific augmentation techniques or attention-based model refinements. Nonetheless, the modular and scalable architecture of the framework facilitates rapid adaptation to additional crops or disease categories, making it highly extensible for real-world smart farming ecosystems.

In conclusion, this federated deep learning system not only demonstrates technical soundness in model training and deployment but also aligns with the overarching goals of sustainable agriculture promoting early disease detection, empowering local decision-making, and preserving data privacy. It establishes a strong foundation for the development of real-time, mobile-integrated diagnostic tools that are scalable, secure, and context-aware across diverse farming environments. To further advance federated learning in agriculture, the following directions are proposed. Vision Transformers (ViTs): Substitute ResNet-50 with ViTs or hybrid CNN-Transformer architectures to more effectively elucidate intricate disease patterns. Attention Mechanisms: Incorporate spatial-channel attention to enhance differentiation among visually analogous diseases (e.g., Apple Scab versus Black Rot). Personalized Federated Learning (FL): Customize global models to accommodate individual agricultural operations through client-specific layers or hyperparameters. Semi-Supervised Federated Learning (FL): Utilize unlabeled on-farm data through contrastive learning or pseudo-labeling methodologies to alleviate class imbalance. Secure Aggregation (SecAgg): Employ cryptographic methodologies (e.g., homomorphic encryption) to obstruct inference attacks during the process of model updates. Federated Differential Privacy (DP): Fine-tune noise injection techniques to reconcile privacy assurances with the utility of the model.

Integration of TinyML: Implement lightweight Federated Learning (FL) models (e.g., MobileNetV3, EfficientNet-Lite) on edge computing devices (such as drones and Internet of Things (IoT) sensors) to facilitate real-time detection of diseases. Blockchain for Enhanced Auditability: Document FL transactions (including model updates and data contributions) to maintain a high level of transparency and foster trust among various stakeholders. Fusion of Multi-Modal Data: Integrate image data with meteorological information, soil sensor readings, and satellite imagery within FL frameworks to conduct a comprehensive analysis of crop health. Federated Reinforcement Learning (FRL): Enhance decision-making processes at the farm level (such as irrigation strategies and pesticide applications) through the utilization of distributed reinforcement learning agents. Interoperable Frameworks for Federated Learning: Establish open standards for FL applications in agriculture to promote collaborative efforts across international borders (for instance, initiatives led by the Food and Agriculture Organization (FAO)). Federated Benchmarking: Develop standardized datasets (such as "FL-PlantVillage") to facilitate a fair comparison of various FL methodologies.

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